Foundations III and IV

Section A : Fill in the blanks

1. A _____ structure allows a program to choose among alternative execution paths.
2. The statement used to execute a block of code only when a condition is true is _____.
3. The Modern C++ feature introduced in C++17 that allows variable declaration inside an if statement is called _____.
4. The _____ statement is commonly used when selecting among multiple constant values.
5. A _____ loop checks its condition before executing the loop body.
6. A _____ loop executes its body at least once before checking the condition.
7. The Modern C++ loop that directly iterates through elements of a container is called a _____ loop.
8. The keyword _____ immediately terminates a loop.
9. The keyword _____ skips the remainder of the current iteration.
10. A _____ is another name for an existing variable.
11. A _____ stores the memory address of another variable.
12. The Modern C++ container that has a fixed size known at compile time is _____.
13. The Modern C++ container that can grow and shrink dynamically is _____.
14. Memory is dynamically allocated using the keyword _____.
15. Memory allocated dynamically is released using _____.
16. A pointer that refers to memory that has already been released is called a _____ pointer.
17. Modern C++ uses _____ pointers to manage memory automatically.

Section B: Explain the differences between

1. A reference and a pointer
2. The std::array and std::vector
3. Passing an argument by value and passing an argument by reference

Getting organised I – Structure within Structure, Functions

Section B : Fill in the blanks

1. _____ is a C++ design principle that acquires resources in a constructor and releases them in a destructor.
2. A _____ is a data type defined by its operations and behaviour rather than its implementation details.
3. A _____ is a data type defined by its operations and behaviour rather than its implementation details.
4. A _____ allows multiple members to share the same memory location.
5. A _____ is a user-defined type that represents a set of named constants.
6. The Modern C++ alternative to traditional `enum` is _____.

7. _____ creates an alias or nickname for an existing type.
8. The Modern C++ alternative to `typedef` is _____.
9. _____ generates a seed value for random number generation.
10. _____ is a pseudo-random number generator commonly used in Modern C++.
11. _____ is used to read data from a file.
12. _____ is used to write data to a file.
13. _____ is used for both reading and writing files.
14. _____ is used to report error messages to the standard error stream.
15. _____ reads an entire line of text, including spaces.
16. The Modern C++ replacement for `rand()` and `srand()` is the _____ library.
17. The Modern C++ replacement for a traditional union is _____.
18. The Modern C++ replacement for manual memory management using `new` and `delete` is _____.
19. The destructor of an object is automatically called when the object goes out of _____.
20. In a structure, members are _____ by default.

Getting organised II: File Handling

LAQ 1 : *Write a program that creates a file named student.txt and stores the following information:*

Name: Ali
Id: 1001
GPA: 3.85

LAQ 2 : *Write a program that reads the data from product.txt and displays in the sample output as shown:*

Product Id: P001
Product Description: Cupboard
Price RM100
Assume there are rows of data stored in product.txt